representation. What are the possible compositional models you can apply to generate the sentence vector?

10. In information retrieval, you are given a set of documents and a query. You are asked to retrieve the set of documents that are most relevant to the query. Many of the common information retrieval techniques are built around the measure of TF-IDF, where TF stands for the term frequency, i.e., the number of times a query keyword appears in a document. It is obvious that one can compute the term frequency for the query terms, and return a ranked list of documents in which the query terms appear most frequently. If this suffices to return the most relevant documents, why do we need the measure of the inverse document frequency (IDF)? Can you illustrate with an example why IDF is needed in information retrieval?
11. You have been given the problem of sentiment analysis, where, given a sentence, you need to classify it as bearing positive, negative or neutral sentiments. You have been given large quantities of labelled data of example sentences, with their target sentiment polarity annotated. Given that there can be words in the test input sentences that are not present in the training set examples, how would your approach handle the 'out of vocabulary' words in the test sentences? If you decide to just ignore the 'out of vocabulary' words, what would be the impact on the test set classification accuracy? Can you devise a better approach to handling this issue?
12. In machine learning, the metrics used in comparing the classifier's performance among different models are precision, recall, area under the Receiver Operating Characteristic (ROC) curve, and the confusion matrix. Can you explain each of these terms?
13. You are asked to build a system that can do 'parts of speech' tagging for a given sentence. While 'parts of speech' tagging assigns independent labels for each of the words in the sentence, it uses the sequence of words it has seen in the context to decide on the current label. For example, consider the sentence: "Robert found the saw on the table and was taking it outside when he saw the snake." The first occurrence of 'saw' in the above sentence is a verb phrase whereas the second occurrence of it is a noun phrase. By looking at the sequence of words occurring around the specific word, a POS tagging system can distinguish between the tags it should assign for each of the two occurrences of the word 'saw'. Hence, POS tagging is an example of sequence labelling. Two of the important algorithms for sequence labelling are Hidden Markov Models and Conditional Random Fields. Can you explain each of them?
14. If you are asked to do sequence labelling with a deep neural network, what type of neural network would you employ? Explain the reason behind your choice.
15. Given an array of N integers, you are asked to find out whether the array is an arithmetic series. An arithmetic series is a sequence of numbers where there is a constant difference between any two consecutive numbers of the series. What is the complexity of your function? Now, you are given a modified problem for which your task is to convert a given array into an arithmetic sequence by dropping off those array elements that prevent it from being an arithmetic sequence. How would you solve this problem? What is the time complexity of your function?
16. You are given a list of N integers, where the numbers can only be positive. You are now being asked to write code to create two lists of size N/2 such that the sum of the elements of the two sub-lists are equal, if at all it is possible to create such sub-lists. Your function should return the sum of the sub-list elements if it is possible to split the original list into two such sub-lists or return zero if that is not possible. What is the time complexity of your function? (Note that you can move the elements of the list.) You are now told that the original list can contain either positive or negative integers and again told to solve the above problem. What are the changes you would make to your earlier solution?
17. Given that the same words can have different meanings (this is known as polysemy), how do information retrieval systems figure out what is the specific information that is being searched for? For example, consider a user who searches for 'windows'. How would the search engine determine whether he/she is looking for Microsoft Windows or glass windows? Blindly displaying search results for both meanings is likely to impact the user experience. If you are the designer of the search engine, explain how you would approach this problem? Consider the related problem. When the user types in the search query as 'automobile', how does the search engine include Web pages that talk about cars as well?
18. What is an ensemble of classifiers? We find that, in many cases, an ensemble of classifiers achieves a better performance compared to any individual classifier model. Can you explain the intuition behind the performance improvements achieved by ensemble classifiers?
19. A well-known phrase used in machine learning and statistics is 'correlation is not the same as causation'. Can you give an example to illustrate this statement?
20. There are three kinds of analytics – namely, prescriptive analytics, descriptive analytics and predictive analytics. Can you explain each of these terms and how they are different from each other?

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Wishing all our readers happy coding until next month! END

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The author is an expert in systems software and is currently working as a research scientist at Conduent Labs India (formerly Xerox India Research Centre). Her interests include compilers, programming languages, file systems and natural language processing. If you are preparing for systems software interviews, you may find it useful to visit Sandya's LinkedIn group Computer Science Interview Training India at *http://www.linkedin.com/groups?home=&gid=2339182*.